



Reciprocal longitudinal associations of supportive workplace relationships with depressive symptoms and self-rated health: A study of Korean women

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ABSTRACT

Background: Supportive relationships in the workplace are important determinants of an individual's psychological health, buffering work stress. This study aimed to estimate the bidirectional association of workplace relationships with depressive symptoms and self-rated health in female workers.

Methods: Based on the nationally representative biannual survey data from 2014 (T1) to 2020 (T4) (N = 2556), we used a random intercept cross-lagged panel model (RI-CLPM) to estimate autoregressive and cross-lagged bidirectional associations between workplace relationships and health variables.

Results: Regarding satisfactory workplace relationships and depressive symptoms, a negative correlation was observed at the between-person level. At the within-person level, the cross-lagged path showed that an increase in satisfaction with workplace relationships was related to a subsequent decrease in depressive (T1 to T2: $\beta = -0.039$; B = -0.253 , P = 0.017). Further, an increase in depressive symptoms was related to subsequent relationship deterioration in the workplace (T1 to T2: $\beta = -0.049$, B = -0.007 , P = 0.003). Regarding workplace relationships and self-rated health, a positive correlation was observed at the between-person level. At the within-person level, there was no significant within-person level association between WPR and self-rated health, and the effect sizes were small.

Conclusions: This study suggests a bidirectional association between workplace relationships and depressive symptoms. Therefore, it is important to implement intervention programs that promote workplace interpersonal relationships and workers' mental health.

1. Introduction

Work is a major determinant of an individual's mental health. Workers are exposed to various psychosocial environments in the workplace, which could have protective or negative effects on their mental health. Previous studies have consistently demonstrated that psychosocial hazards are associated with mental health problems, such as depressive and anxiety symptoms and sleep disorders (Niedhammer et al., 2021; van der Molen et al., 2020). For example, work-related psychosocial risk factors, such as high job demands with low rewards (Lim et al., 2021; Nigatu and Wang, 2018), organizational injustice (Hayashi et al., 2019), and experiencing a workplace violence (Lee et al., 2022), have negative impact on worker's mental health. Conversely,

positive organizational culture and leadership have the potential to reduce employee burnout and contribute to improved mental health (Sorengaard and Langvik, 2022). Therefore, for stress management and health promotion of workers, it is crucial to understand the various psychosocial working environments that can affect their mental health (Tetrick and Winslow, 2015).

1.1. Workplace relationship and mental health

In recent decades, research interest in the work environment has gradually shifted from physical and toxicological risk factors to psychosocial ones (Nieuwenhuijsen et al., 2010; Rugulies, 2019; Stansfeld and Candy, 2006; van der Molen et al., 2020). In particular, a supportive

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relationship with co-workers is a key element in the psychosocial work environment, buffering work stress and contributing to uplifting work performance (Kundi et al., 2022). Workers with supportive relationship at the workplace are less likely to develop mental health problems, such as depression (Park et al., 2004). Previous studies have suggested that supportive workplace relationships can have a protective effect on mental health, particularly for workers who have experienced trauma in the workplace, by reducing the risk of developing post-traumatic stress disorder (Oginska-Bulik, 2013; Pietrzak et al., 2014). In contrast, destructive workplace relationships, such as workplace bullying or verbal abuse, are closely related to depression or anxiety (K. Kim et al., 2020b; Lee et al., 2022). Similarly, interpersonal conflicts at the workplace can be one of major contributors for work-related stressors (Ilies et al., 2011). However, a recent study by Zuelke found no significant relationship between interpersonal conflicts and depressive symptoms among workers, suggesting that the relationship between them may be confounded by personal traits (e.g., neuroticism) (Zuelke et al., 2020).

1.2. Gender difference in the association between workplace relationship and mental health

While social support can profoundly affect the psychological health of both men and women, research suggests that it has a greater impact on mental health in women than in men (Milner et al., 2016). Similar findings have been observed in the occupational context, with Fuhrer et al. (1999) suggesting that the psychological impact of workplace relationship on mental health is greater for women than for men (Fuhrer et al., 1999). However, with regard to reverse causation, Rosander et al. found that mental health problems predicted future workplace bullying only among male workers (Rosander et al., 2020). Women experience different characteristics of workplace interrelationships than men do, sometimes becoming victims of sexual harassment or discrimination (G. Kim et al., 2020a; Rosander et al., 2020). The findings of the European Working Conditions Survey have revealed that a higher proportion of women (4.5%) compared to men (3.9%) are subject to workplace bullying or harassment (Hodgins, 2014). Therefore, from the perspective of gender, examining the association between workplace social relationships and the health of women is important for promoting women's mental health and economic participation.

1.3. Limitation of previous studies

Although several studies have found an association between workplace relationships and worker health, certain limitations have been pointed out, as most of them are based on a cross-sectional design. First, the observed correlation might have resulted from the confounding effect of an individual's personality or traits. For example, workers with neuroticism tend to experience negative workplace relationships and report poor psychological health simultaneously, while the extroverted do the opposite (Deniz and Ertosun, 2010; Malouff et al., 2005; Pallesen et al., 2017). Therefore, differences in personality or traits between individuals can overestimate the association between workplace relationships and health. Second, reverse causation should be considered; that is, mental health problems or poor physical conditions can also cause workplace relationships to deteriorate (Carr and Namkung, 2021; Rosander et al., 2020). Workers who exhibit a high degree of negative emotions may be perceived as difficult to communicate with, making them more susceptible to becoming targets of workplace bullying and aggression (Aquino and Thau, 2009; Rosander, 2021). The association between workplace relationships and depressive symptoms may be bidirectional and reciprocal, as each factor can influence the other over time in a longitudinal perspective (Finne et al., 2011; Rosander et al., 2020). Nevertheless, little is known in the current literature about how workers' depressive symptoms and self-rated health affect workplace relationships, and most studies have treated worker health as an outcome.

1.4. Present study

To overcome limitations, we employed a random intercept cross-lagged panel model (RI-CLPM) to explore the longitudinal association of workplace relationships with workers' depressive symptoms and self-rated health. Using the RI-CLPM enables us to explore bidirectional associations between workplace relationships and health, while also disentangling between-person differences and within-person effects (Hamaker et al., 2015; Mulder and Hamaker, 2021). By examining cross-lagged associations, our findings contribute to the literature by offering new insights into the dynamic association between workplace social relationships and worker health.

2. Methods

2.1. Study sample

Our dataset was obtained from the 5th to 8th wave (2014–2020) of the Korean Longitudinal Survey of Women and Families (KLoWF), a biannually conducted nationwide survey by the Korean Women's Development Institute. Since 2007, KLoWF researchers have followed up a national representative sample comprising 9997 women aged 19–64 years to investigate Korean women's economic activities, sexual and family related values, and health. As the questionnaire on depressive symptoms has been included since 2014, we conducted our analysis based on survey participants from 2014 (Time 1, T1) to 2020 (Time 4, T4).

The KLoWF sampled study participants using probability proportional to size sampling based on the Korean National Population and Housing Census conducted in 2005. Two-stage stratified sampling was used to select study participants, where the enumeration districts were considered as the primary sampling unit and the households living in the selected survey area as the secondary sampling unit. Therefore, the KLoWF survey participants represented all adult women in Korea. The survey was conducted via face-to-face interviews with professional interviewers.

Of the 7745 survey participants included in 2014, 3480 women who did not have a job were excluded. Next, we excluded 29 participants with missing values in their baseline characteristics (age, educational level, household income, occupation, and employment arrangement), leaving 4236 remaining participants. Our main analyses were based on the complete cases that included 2556 survey participants who have consistently participated in surveys from T1 to T4.

2.2. Data availability and ethics statement

The raw KLoWF data are publicly available online. Interested researchers can check the KLoWF website for detailed information on its purpose, design, sampling, and survey items (<https://klof.kwdi.re.kr/portal/eng/mainPage.do>). The data did not include personal information of the survey participants, and were anonymized by KLoWF researchers before publication. This study was reviewed and approved by the Institutional Review Board of Yonsei Health System (4-2022-1065).

2.3. Study variables

2.3.1. Workplace relationships

Workplace relationship was assessed using the following question: "How much are you satisfied or dissatisfied with communication and relationships in the workplace?" Responses were rated on a 5-point Likert scale (1 = very dissatisfied, 2 = somewhat dissatisfied, 3 = neither satisfied nor dissatisfied, 4 = slightly satisfied, 5 = very satisfied), with higher scores indicating more satisfactory workplace relationships.

2.3.2. Depressive symptoms

Depressive symptoms were measured using the 10-item Centre for Epidemiological Studies Depression Scale (CES-D-10). The Korean version of the CES-D-10 has been shown to possess high levels of reliability and validity in the Korean population, as established in the previous study (Shin, 2011). The CES-D-10 comprises 10 items related to depressive symptoms, and responses are based on an integer between 0 and 3. Therefore, the total score for individual depressive symptoms ranges between 0 and 30, with higher scores indicating higher levels of depressive symptoms. Cronbach's alpha for each year ranged from 0.85 to 0.88.

2.3.3. Self-rated health

Self-rated health was assessed using the following question: "What do you think about your current health condition?" Responses were rated on a 5-point Likert scale (1 = very bad, 2 = somewhat bad, 3 = neither bad nor good, 4 = somewhat good, 5 = very good).

2.3.4. Other covariates

We controlled for the following covariates at T1 in the RI-CLPM model: Age was categorized as <40 years, 40–49 years, 50–59 years, and ≥60 years. Education was categorized as having completed middle school or below, high school, or college or above. Household income was classified into quartiles. Occupation was categorized as blue collar, service or sales worker, or white collar. Employment arrangement was categorized as permanent employment, fixed-term employment, or self-employment or others. Additionally, self-rated health was controlled in the analysis of workplace relationship and depressive symptoms. Similarly, depressive symptoms were controlled in the analysis of workplace relationship and self-rated health.

2.4. Statistical analysis

First, we presented the baseline characteristics of the study sample. For preliminary analysis, the zero-order correlation (Spearman's rank correlation coefficient) between the main variables across all four waves

was calculated. We then calculated the intraclass correlations (ICCs) of workplace relationships, depressive symptoms, and self-rated health. ICC represents the proportion of the between-person variance to the total variance.

To estimate the bidirectional, longitudinal relationship between workplace relationships and health variables, we employed the cross-lagged panel models (CLPMs). Using the CLPM, researchers can estimate autoregressive and cross-lagged bidirectional associations between the variables of interest. However, the traditional CLPM has been criticized for its limitation in that it may not represent true within-person causal influences (Hamaker et al., 2015). Contrarily, the alternative RI-CLPM has the advantage of separating within-person influences and trait-like, between-person differences by adding a random intercept to the structural equation model (Mulder and Hamaker, 2021). Therefore, using the RI-CLPM, we could estimate how time-stable differences in workplace relationships are related to time-stable differences in each health variable between persons, and how fluctuations in within-person workplace relationships are related to fluctuations in each health variable. Fig. 1 depicts the RI-CLPM, which represents the association between workplace relationship and depressive symptoms, and Fig. 2 depicts the other one, which represents the association between workplace relationship and self-rated health.

Next, for the model-building procedure, we sequentially created different models, in which the autoregressive and cross-lagged path coefficients were constrained to be equal across the waves. Covariates were added as predictors of each intercept. The optimal model structure was determined by comparing the model fit indices.

All parameters in the RI-CLPMs were estimated based on the maximum likelihood robust (MLR) method to deal with the non-normal distribution of the study variables. The goodness of fit for the RI-CLPMs were evaluated using chi-square statistics (χ^2), comparative fit index (CFI), Tucker-Lewis index (TLI), and root mean square error of approximation (RMSEA). Good model fit was defined as CFI ≥ 0.95, TLI ≥ 0.95, and RMSEA ≤ 0.06 (Hu and Bentler, 1999). For model comparison, $\Delta\text{CFI} \geq -0.01$ and $\Delta\text{RMSEA} \leq 0.015$ were considered acceptable changes in the model fit (Cheung and Rensvold, 2002).

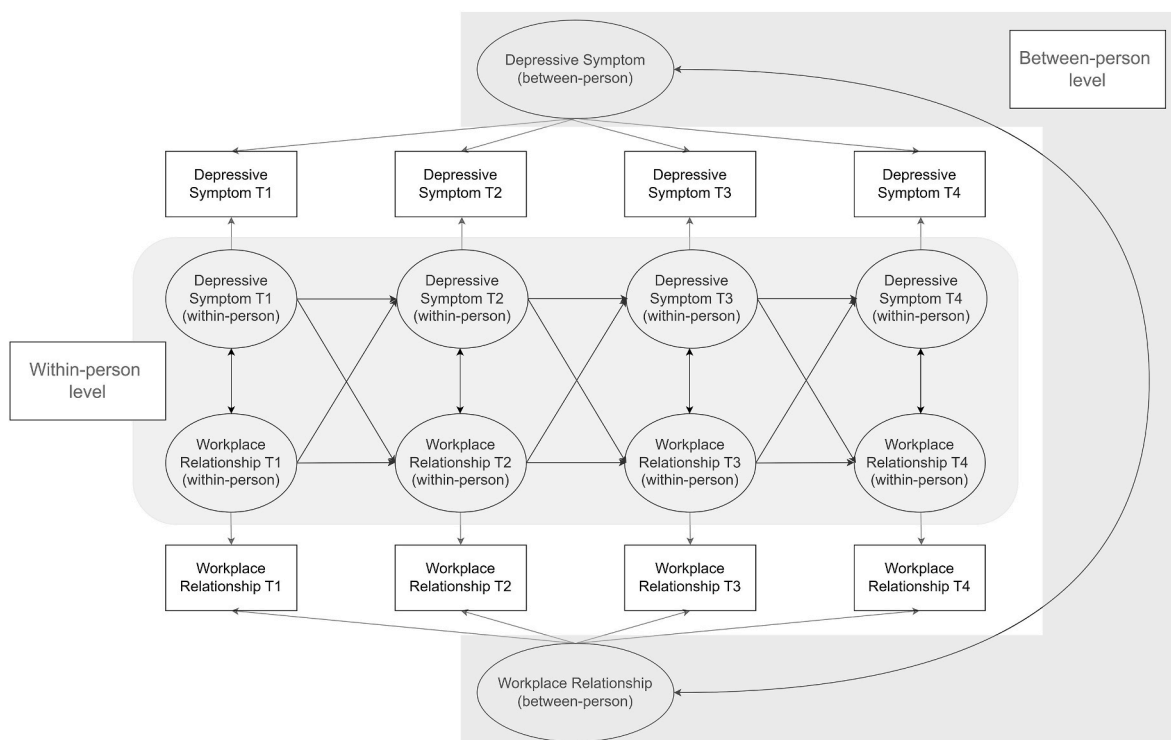


Fig. 1. Random intercept cross-lagged panel model of the association between workplace relationships and depressive symptoms.

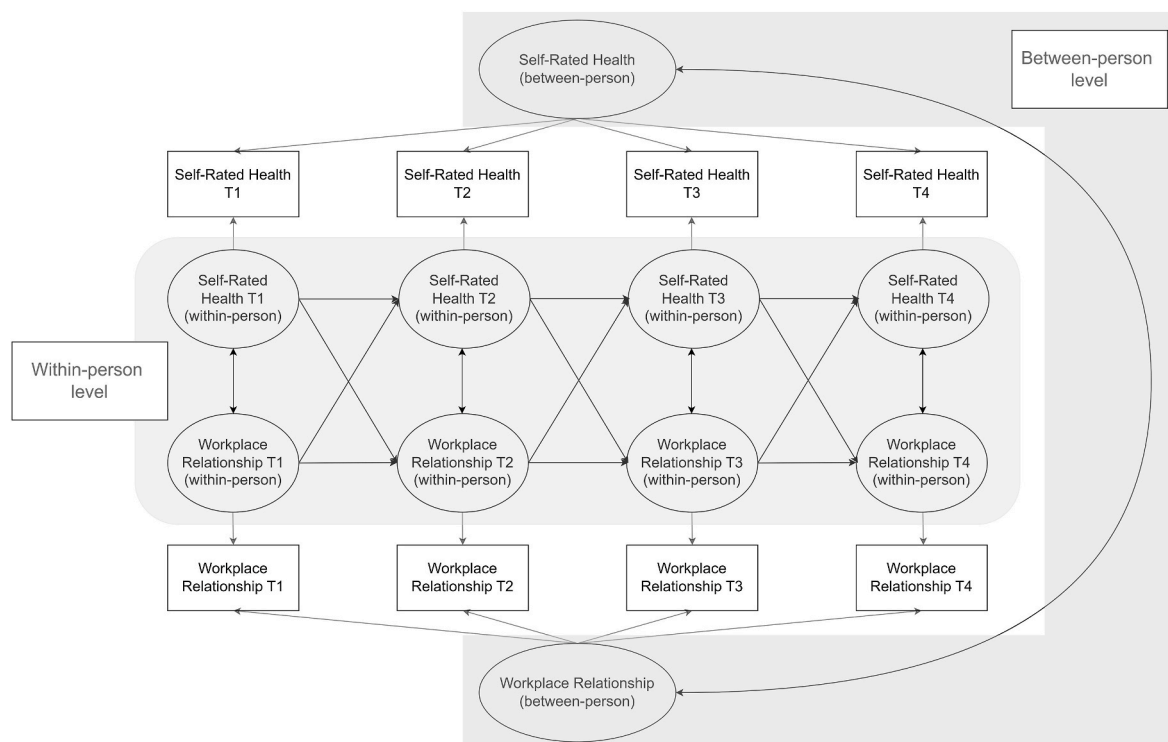


Fig. 2. Random intercept cross-lagged panel model of the association between workplace relationships and self-rated health.

Finally, participant attrition occurred due to loss to follow-up or retirement. The proportion of participants with missing values for at least one or more variables was 17.5% at T2, 24.5% at T3, and 32.2% at T4; the number of complete cases was 2556 (60.3%). A dropout analysis was conducted to compare the characteristics of the study samples according to the participation pattern. First, we performed main analyses based on complete cases ($N = 2556$). We then employed a FIML (Full Information Maximum Likelihood) estimation to fully use cases with missing values. The FIML estimates unbiased parameters using “full information” on all observed variables according to each case (Enders, 2010). Additional information on the FIML approach used in this study can be found in the supplementary material (see Supplementary Details). The results based on FIML estimation are also presented in supplementary materials ($N = 4236$).

Statistical significance was defined as a p -value of less than 0.05. All statistical analyses were performed using R (version 4.2.2; R Foundation for Statistical Computing, Vienna, Austria). The “lavaan” package was used to fit the RI-CLPMs (Rosseel, 2012).

3. Results

3.1. Preliminary analysis

A total of 2556 women were included in the current study. The baseline characteristics of the study participants are presented in Table S1. Dropout analysis indicated that those with missing values were more likely to be aged under 40 years, be white collars, have higher educational attainment, and have higher income levels. There were no significant differences in depressive symptoms and workplace relationship between complete and missing cases. However, those with missing cases had better self-rated health (Table S2). Table 1 presents the zero-order correlations between the study variables. Depressive symptoms were negatively correlated with self-rated health and workplace relationships. In contrast, a positive correlation was observed between self-rated health and workplace relationships. The ICCs were 0.366 (depressive symptoms), 0.453 (self-rated health), and 0.323 (workplace relationship), indicating that the between-person variance explained 36.6%, 45.3%, and 32.3% of the total variance in depressive symptoms,

Table 1

Zero-order correlation between depressive symptoms, self-rated health, and workplace relationship across all waves (SD: standard deviation; DPR: depressive symptoms; SRH: self-rated health; WPR: workplace relationships).

Variables	Median	Q1–Q3	1	2	3	4	5	6	7	8	9	10	11	12
Time 1														
1. DPR	3	0–7	1											
2. SRH	4	3–4	–0.20	1										
3. WPR	3	3–4	–0.15	0.22	1									
Time 2														
4. DPR	3	0–6	0.43	–0.17	–0.15	1								
5. SRH	4	3–4	–0.18	0.45	0.21	–0.29	1							
6. WPR	3	3–4	–0.13	0.18	0.38	–0.17	0.21	1						
Time 3														
7. DPR	2	0–7	0.32	–0.12	–0.10	0.41	–0.20	–0.11	1					
8. SRH	4	3–4	–0.17	0.40	0.20	–0.20	0.51	0.20	–0.24	1				
9. WPR	3	3–4	–0.091	0.20	0.29	–0.10	0.21	0.38	–0.13	0.29	1			
Time 4														
10. DPR	3	1–7	0.30	–0.21	–0.14	0.32	–0.30	–0.14	0.42	–0.27	–0.17	1		
11. SRH	4	3–4	–0.17	0.36	0.19	–0.20	0.42	0.18	–0.21	0.48	0.24	–0.34	1	
12. WPR	3	3–4	–0.14	0.18	0.27	–0.13	0.18	0.32	–0.17	0.22	0.37	–0.22	0.27	1

*All correlations were significant at $P < 0.001$.

self-rated health, and workplace relationships, respectively.

3.2. Workplace relationships and depressive symptoms

The model fit indices of the RI-CLPMs for the relationship between workplace relationships and depressive symptoms are shown in Table 2. All RI-CLPMs had a good model fit, and the model fit indices did not change significantly when the path coefficients were constrained across the waves. The estimated coefficients are presented in Table 3. A negative correlation was observed at the between-person level (standardized $\beta = -0.187$, unstandardized $B = -0.119$, $P < 0.001$). At the within-person level, the autoregressive paths showed that depressive symptoms in the previous wave were related to those of the subsequent wave (T1 to T2: $\beta = 0.186$, $B = 0.178$, $P < 0.001$). Similarly, workplace relationship in the previous wave was related to that of the subsequent wave (T1 to T2: $\beta = 0.133$, $B = 0.133$, $P < 0.001$). The cross-lagged path showed that an increase in satisfaction with workplace relationships was related to a subsequent decrease in depressive symptoms (T1 to T2: $\beta = -0.039$, $B = -0.253$, $P = 0.017$). In addition, an increase in depressive symptoms was related to subsequent relationship deterioration in the workplace (T1 to T2: $\beta = -0.049$, $B = -0.007$, $P = 0.003$).

3.3. Workplace relationships and self-rated health

The model fit indices of the RI-CLPMs for the relationship between workplace relationships and self-rated health are presented in Table 2. All RI-CLPMs had a good model fit, and the model fit indices did not change significantly when the path coefficients were constrained across the waves. The estimated coefficients are presented in Table 4. A positive correlation was observed at the between-person level (standardized $\beta = 0.307$, unstandardized $B = 0.130$, $P < 0.001$). At the within-person level, the autoregressive paths showed that self-rated health in the previous wave were related to that of the subsequent wave (T1 to T2: $\beta = 0.159$, $B = 0.154$, $P < 0.001$). There was weak evidence that workplace relationships were associated with subsequent changes in self-rated health (T1 to T2: $\beta = 0.032$, $B = 0.034$, $P = 0.052$), and vice versa (T1 to T2: $\beta = 0.028$, $B = 0.026$, $P = 0.119$).

3.4. Sensitivity analyses

To confirm the robustness of the current findings, we performed sensitivity analyses based on the FIML method ($N = 4236$). Tables S3 and S4 present the results of sensitivity analyses. Both RI-CLPMs showed good model fit indices. In the sensitivity analyses, the longitudinal,

Table 3

Estimated coefficients for the model of workplace relationships and depressive symptoms (S.E: standard error; DPR: depressive symptoms; WPR: workplace relationships).

Paths	Unstandardized coefficients	S.E	Standardized coefficients	P value
Between-person level				
Correlation				
DPR $\leftrightarrow$ WPR	-0.119	0.036	-0.187	<0.001
Within-person level				
Autoregressive paths				
DPR _{T1} $\rightarrow$	0.178	0.022	0.186	<0.001
DPR _{T2} $\rightarrow$	0.178	0.022	0.169	<0.001
DPR _{T3} $\rightarrow$	0.178	0.022	0.176	<0.001
DPR _{T4} $\rightarrow$	0.133	0.021	0.133	<0.001
WPR _{T1} $\rightarrow$	0.133	0.021	0.136	<0.001
WPR _{T2} $\rightarrow$	0.133	0.021	0.127	<0.001
WPR _{T3} $\rightarrow$	0.133	0.021		
WPR _{T4} $\rightarrow$				
Cross-lagged paths				
WPR _{T1} $\rightarrow$	-0.253	0.106	-0.039	0.017
DPR _{T2} $\rightarrow$	-0.253	0.106	-0.037	0.017
DPR _{T3} $\rightarrow$	-0.253	0.106	-0.036	0.017
DPR _{T4} $\rightarrow$	-0.007	0.003	-0.049	0.003
WPR _{T2} $\rightarrow$	-0.007	0.003	-0.048	0.003
DPR _{T3} $\rightarrow$	-0.007	0.003	-0.048	0.003
WPR _{T4} $\rightarrow$				
(Residual) correlations				
DPR _{T1} $\leftrightarrow$	-0.160	0.058	-0.069	0.006
WPR _{T1} $\leftrightarrow$	-0.216	0.055	-0.101	<0.001
DPR _{T2} $\leftrightarrow$	-0.174	0.057	-0.079	0.002
WPR _{T2} $\leftrightarrow$	-0.243	0.058	-0.105	<0.001
DPR _{T3} $\leftrightarrow$				
WPR _{T3} $\leftrightarrow$				
DPR _{T4} $\leftrightarrow$				
WPR _{T4} $\leftrightarrow$				

cross-lagged effects between workplace relationships and depressive symptoms were confirmed. In case of the self-rated health, statistically significant cross-lagged effects between workplace relationships and self-rated health emerged at the within-person level.

Table 2

Fit indices of the RI-CLPM models (df : degrees of freedom; CFI: comparative fit index; TLI: Tucker-Lewis index; RMSEA: root mean square error of approximation).

Models	χ^2	df	CFI	TLI	RMSEA	ΔCFI	$\Delta RMSEA$
RI-CLPM models for the relationship between workplace relationship (WPR) and depressive symptom (DPR)							
Model 1 Baseline model	166.188	87.000	0.980	0.969	0.019	–	–
Model 2 Model 1 + autoregressive path constrained (1) (DPR $\rightarrow$ DPR)	165.424	89.000	0.980	0.971	0.018	0.000	-0.001
Model 3 Model 2 + autoregressive path constrained (2) (WPR $\rightarrow$ WPR)	164.660	91.000	0.981	0.973	0.018	0.001	0.000
Model 4 Model 3 + cross-lagged path constrained (1) (WPR $\rightarrow$ DPR)	167.101	93.000	0.981	0.973	0.018	0.000	0.000
Model 5 Model 4 + cross-lagged path constrained (2) (DPR $\rightarrow$ WPR)	180.089	95.000	0.978	0.970	0.019	-0.003	0.001
RI-CLPM models for the relationship between workplace relationship (WPR) and self-rated health (SRH)							
Model 1 Baseline model	124.113	87.000	0.993	0.990	0.013	–	–
Model 2 Model 1 + autoregressive path constrained (1) (SRH $\rightarrow$ SRH)	134.294	89.000	0.992	0.988	0.014	-0.001	0.001
Model 3 Model 2 + autoregressive path constrained (2) (WPR $\rightarrow$ WPR)	133.795	91.000	0.992	0.989	0.014	0.000	0.000
Model 4 Model 3 + cross-lagged path constrained (1) (WPR $\rightarrow$ SRH)	135.255	93.000	0.992	0.989	0.013	0.000	-0.001
Model 5 Model 4 + cross-lagged path constrained (2) (SRH $\rightarrow$ WPR)	138.594	95.000	0.992	0.989	0.013	0.000	0.000

Table 4

Estimated coefficients for the model of workplace relationships and self-rated health (S.E.: standard error; SRH: self-rated health; WPR: workplace relationships).

Paths	Unstandardized coefficients	S.E	Standardized coefficients	P value
Between-person level				
Correlation				
SRH ↔ WPR	0.130	0.006	0.307	<0.001
Within-person level				
Autoregressive paths				
SRH _{T1} →	0.154	0.022	0.159	<0.001
SRH _{T2} →	0.154	0.022	0.160	<0.001
SRH _{T3} →	0.154	0.022	0.151	<0.001
SRH _{T4} →	0.154	0.022	0.151	<0.001
WPR _{T1} →	0.135	0.021	0.135	<0.001
WPR _{T2} →	0.135	0.021	0.139	<0.001
WPR _{T3} →	0.135	0.021	0.129	<0.001
WPR _{T4} →	0.135	0.021	0.129	<0.001
Cross-lagged paths				
WPR _{T1} →	0.034	0.018	0.032	0.052
SRH _{T2} →	0.034	0.018	0.034	0.052
SRH _{T3} →	0.034	0.018	0.032	0.052
SRH _{T4} →	0.026	0.017	0.028	0.119
WPR _{T2} →	0.026	0.017	0.028	0.119
SRH _{T3} →	0.026	0.017	0.026	0.119
WPR _{T4} →	0.026	0.017	0.026	0.119
(Residual) correlations				
SRH _{T1} ↔	0.014	0.009	0.037	0.128
WPR _{T1} ↔	0.017	0.010	0.048	0.082
SRH _{T2} ↔	0.017	0.010	0.048	0.082
WPR _{T2} ↔	0.044	0.009	0.132	<0.001
SRH _{T3} ↔	0.044	0.009	0.132	<0.001
WPR _{T3} ↔	0.044	0.009	0.124	<0.001
SRH _{T4} ↔	0.044	0.009	0.124	<0.001
WPR _{T4} ↔	0.044	0.009	0.124	<0.001

4. Discussion

The main result of our study was that positive workplace relationships were not only closely related to the subsequent changes in mental health, but also that an increase in depressive symptoms were related to the subsequent deterioration of workplace relationships among women. In particular, by separating between-person and within-person variance, we were able to confirm the longitudinal cross-lagged association between workplace relationships and depressive symptoms at the within-person level. To the best of our knowledge, this is the first study to investigate the reciprocal association between workplace relationships and health based on the RI-CLPM.

The estimated coefficients of the cross-lagged paths were relatively small. However, since the auto-regressive effect accounts for a large part of the variance in RI-CLPM, effect size conventions suggested for interpreting correlation coefficients, such as 0.10 indicating a small effect, do not apply to cross-lagged effects (Adachi and Willoughby, 2015). A recent study by Orth et al. suggested that for RI-CLPM, standardized coefficients of 0.03, 0.07, or 0.12 should be interpreted as a small, medium, and large effect, respectively (Orth et al., 2022). Based on their findings, the effect of depressive symptoms on workplace relationships was small to medium, and the effect of workplace relationships on depressive symptoms was small. Additionally, the estimated coefficients of cross-lagged effects between self-rated health and workplace relationship corresponded to small effects.

Our findings are consistent with prior studies which maintain that

workplace relationships are closely associated with the psychological health of workers. In previous studies, workplace support has consistently been associated with better mental health outcomes (Deming and Vasterling, 2017; Oginska-Bulik, 2013; Pietrzak et al., 2014). In contrast, negative workplace relationships, such as workplace bullying, are related to poor mental health (Lee et al., 2022). In summary, numerous studies on workplace relationships and psychological health have been published thus far, and they have consistently shown a significant correlation. Our study is meaningful in that it further clarifies temporal relationships and verifies these associations at the within-person level. In contrast, contradictory to the results of the previous cross-sectional study (Oxenstierna et al., 2011), the workplace relationships and self-rated health had no significant temporal associations at the within-person level, with only correlation at the between-person level being identified.

The temporal relationship between negative workplace relationships and subsequent change in depressive symptoms is not inconceivable. According to previous studies, social support plays a crucial role in buffering the effect of job stress on mental health (Jolly et al., 2021; Zhang et al., 2020). Furthermore, physiological mechanisms account for the influence of workplace relationships on depressive symptoms, in which social support can alter the activity of the hypothalamic-pituitary-adrenal (HPA) axis, which is closely related to psychological health (Iob et al., 2018; Monreal et al., 2022; Ramdas et al., 2022). Although social support at the workplace was not addressed in Iob's study, it is plausible that workplace relationships may also affect psychological health through the HPA axis, considering that workers spend a substantial amount of time at work.

Our study also revealed a within-person-level association between an increase in depressive symptoms and subsequent negative workplace relationships. Despite many studies on the positive or negative effects of social support on mental health in the workplace setting, health has mostly been treated as a result of workplace relationships rather than a predictor. Nevertheless, depressive symptoms are well-known factors in worsening social functions (Oliveri et al., 2020; Santini et al., 2020). Depression symptoms interfere with active social participation and communication; and sometimes, stigma for depression may contribute to social withdrawal (Yang et al., 2020). Additionally, an increase in depressing emotions can induce individuals to evaluate their social relations more negatively in the workplace (Kompier and Taris, 2011). In fact, some preceding studies have revealed that workers with mental health problems are more likely to experience destructive workplace relationships, such as bullying (Finne et al., 2011; Rosander et al., 2020). In line with these previous studies, the results of our study indicate that depressive symptoms are associated with subsequent deterioration of workplace relationships.

From a gender perspective, our findings deepen our understanding of the association between women's mental health and social relations in the workplace. According to a previous meta-analysis, women perceive negative workplace relationships more sensitively than men, although there is no significant difference in their consequences between the sexes (Rosander et al., 2020). However, gender differences have been reported in the reverse causation; that is, mental health problems at baseline predicted negative workplace relationships only for men in previous studies (Einarsen and Nielsen, 2015; Rosander et al., 2020). However, the results of our study demonstrate that poor mental health could be associated with the subsequent deterioration of workplace relationships among female workers, which contradicts the previous findings that poor mental health is associated with a subsequent deterioration of workplace relationship only for male workers. The disparity in findings between our study and previous ones could be attributed to the use of different outcome variables. Specifically, previous studies have primarily examined the incidence of workplace bullying as an outcome, while our study focused on workplace relationships. Given that workplace bullying is a more severe form of mistreatment and less frequent than a decline in interpersonal relationships, it is possible that the

association between poor mental health and workplace bullying was not statistically significant in previous studies.

Finally, the bidirectional association between workplace relationships and depressive symptoms suggests a vicious cycle. In other words, negative workplace relationships are associated with an increase in depressive symptoms, which are also associated with the subsequent deterioration of workplace relationships. This concurrent status of poor mental health and dissatisfaction with work can eventually contribute to job loss (D'Angelo et al., 2021). Therefore, our study emphasizes the need for active intervention in this vicious circle to protect women's economic participation and psychological health. Previous studies have consistently demonstrated the importance of promoting mental health and positive workplace relationships to improve employee performance and organizational productivity (de Oliveira et al., 2023; Omunakwe et al., 2018). Therefore, it is essential for organizations to prioritize the cultivation of a supportive work environment that fosters positive workplace relationships.

4.1. Limitations

The present study had several notable limitations. First, the main analysis was based on the complete case analysis, which assumes missing data to be completely at random (MCAR). However, given that participants with a younger age and higher socio-economic status were more likely to have missing data, it is possible that the MCAR assumption may not be met. Nevertheless, a sensitivity analysis based on the FIML approach was conducted, and the findings showed similar associations between workplace relationship and the outcomes as observed in the main analysis. Although FIML has been recommended for calculating unbiased estimations (Graham, 2009), there is a possibility that the observed associations might be affected by attrition samples. Next, since we only investigated a female sample in Korea, it is necessary to verify whether our conclusions are maintained in men or in other cultural contexts (e.g., Western countries). Third, our measure of workplace relationship was based on single scale, which could limit our results from reflecting multidimensional aspects of the workplace relationship. Fourth, the RI-CLPM has a limited potential to adjust for the time-varying covariates, and there is a possibility that unobserved confounders can affect relationships between variables. Fifth, since all of the variables used in this study were measured on self-reported questionnaires, measurement errors and common-method bias should be considered. Therefore, more objective measurements should be considered in prospective studies. Sixth, as with most studies conducted in organizational settings, healthy worker survivor bias should be considered. Especially, those who had better physical and psychological health were more likely to be selected as study sample. Finally, the estimated cross-lagged effects between workplace relationship and depressive symptoms and self-rated health do not necessarily represent the causal relationship between variables, since our analyses were based on an observational study design and there is a possibility that unobserved confounders could affect our estimations.

Despite these limitations, our study had several notable strengths. First, the results were drawn from a nationally representative sample. Therefore, this study has strengths in terms of generalization. Second, we contribute to the literature by exploring the longitudinal, bidirectional relationship between workplace relationships and health variables. This is especially important in that most of the preceding studies were cross-sectional studies that did not consider longitudinal associations or confounding effects of individual personality or traits and simultaneity bias.

5. Conclusions

Overall, this study suggests a reciprocal, bidirectional association between workplace relationships and depressive symptoms. We observed weak evidence of within-person relationship between

workplace relationship and self-rated health, but the effect sizes were small. Our findings provide evidence for implementing effective intervention programs to promote workplace interpersonal relationships and the mental health of workers.

Author statement

Author Contributions: Conceptualization, S.-U.B. and J.-U.W.; methodology, S.-U.B.; formal analysis, S.-U.B.; writing—original draft preparation, S.-U.B.; writing—review and editing, J.-H.Y. and J.-U.W.; visualization, S.-U.B.; supervision, J.-H.Y. and J.-U.W.; All authors have read and agreed to the published version of the manuscript.

Data availability

The raw data are publicly available online. (<https://klowf.kwdi.re.kr/portal/eng/mainPage.do>).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.socscimed.2023.116176>.

References

- Adachi, P., Willoughby, T., 2015. Interpreting effect sizes when controlling for stability effects in longitudinal autoregressive models: implications for psychological science. *Eur. J. Dev. Psychol.* 12, 116–128.
- Aquino, K., Thau, S., 2009. Workplace victimization: aggression from the target's perspective. *Annu. Rev. Psychol.* 60, 717–741.
- Carr, D., Namkung, E.H., 2021. Physical disability at work: how functional limitation affects perceived discrimination and interpersonal relationships in the workplace. *J. Health Soc. Behav.* 62, 545–561.
- Cheung, G.W., Rensvold, R.B., 2002. Evaluating goodness-of-fit indexes for testing measurement invariance. *Struct. Equ. Model.* 9, 233–255.
- D'Angelo, S., Syddall, H., Ntani, G., Harris, E.C., Linaker, C., Cooper, C., et al., 2021. How does job dissatisfaction interact with self-rated health in determining the risk of health-related job loss? Prospective findings from the Health and Employment After Fifty (HEAF) study. *Occup. Environ. Med.* 78, 36–42.
- de Oliveira, C., Saka, M., Bone, L., Jacobs, R., 2023. The role of mental health on workplace productivity: a critical review of the literature. *Appl. Health Econ. Health Pol.* 21, 167–193.
- Deming, C.A., Vasterling, J.J., 2017. Workplace social support and behavioral health prior to long-duration spaceflight. *Aerosp. Med. Hum. Perform.* 88, 565–573.
- Deniz, N., Ertosun, O.G., 2010. The relationship between personality and being exposed to workplace bullying or mobbing. *J. Glob. Strateg. Manag.* 7.
- Einarsen, S., Nielsen, M.B., 2015. Workplace bullying as an antecedent of mental health problems: a five-year prospective and representative study. *Int. Arch. Occup. Environ. Health* 88, 131–142.
- Enders, C.K., 2010. *Applied Missing Data Analysis*. Guilford Publications.
- Finne, L.B., Knardahl, S., Lau, B., 2011. Workplace bullying and mental distress - a prospective study of Norwegian employees. *Scand. J. Work. Environ. Health* 37, 276–287.
- Fuhrer, R., Stansfeld, S.A., Chemali, J., Shipley, M.J., 1999. Gender, social relations and mental health: prospective findings from an occupational cohort (Whitehall II study). *Soc. Sci. Med.* 48, 77–87.
- Graham, J.W., 2009. Missing data analysis: making it work in the real world. *Annu. Rev. Psychol.* 60, 549–576.
- Hamaker, E.L., Kuiper, R.M., Grasman, R.P., 2015. A critique of the cross-lagged panel model. *Psychol. Methods* 20, 102–116.
- Hayashi, T., Odagiri, Y., Takamiya, T., Kikuchi, H., Fukushima, N., Inoue, S., 2019. Exposure to organisational injustice and serious psychological distress: longitudinal analysis of details of exposure from a private Japanese company. *BMJ Open* 9, e029556.
- Hodgins, M., 2014. Workplace Mistreatment: Health, Working Environment and Social and Economic Factors. *Health*, 2014.
- Hu, L.T., Bentler, P.M., 1999. Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. *Struct. Equ. Model.* 6, 1–55.
- Iles, R., Johnson, M.D., Judge, T.A., Keeney, J., 2011. A within-individual study of interpersonal conflict as a work stressor: dispositional and situational moderators. *J. Organ. Behav.* 32, 44–64.
- Iob, E., Kirschbaum, C., Steptoe, A., 2018. Positive and negative social support and HPA-axis hyperactivity: evidence from glucocorticoids in human hair. *Psychoneuroendocrinology* 96, 100–108.
- Jolly, P.M., Kong, D.T., Kim, K.Y., 2021. Social support at work: an integrative review. *J. Organ. Behav.* 42, 229–251.

- Kim, G., Kim, J., Lee, S.K., Sim, J., Kim, Y., Yun, B.Y., et al., 2020a. Multidimensional gender discrimination in workplace and depressive symptoms. *PLoS One* 15, e0234415.
- Kim, K., Kim, I., Youn, K., 2020b. A descriptive study of claims for occupational mental disorder: adjustment disorder. *Ann Occup Environ Med* 32, e39.
- Kompier, M.A., Taris, T.W., 2011. Understanding the causal relations between psychosocial factors at work and health - a circular process. *Scand. J. Work. Environ. Health* 37, 259–261.
- Kundi, Y.M., Soomro, S.A., Kamran, M., 2022. Does social support at work enhance subjective career success? The mediating role of relational attachment. *Int. J. Organ. Anal.* 30, 1491–1507.
- Lee, N.R., Kim, S.W., Joo, J.H., Lee, J.H., Lee, J.H., Lee, K.J., 2022. Relationship between workplace violence and work-related depression/anxiety, separating the types of perpetrators: a cross-sectional study using data from the fourth and fifth Korean Working Conditions Surveys (KWCS). *Ann Occup Environ Med* 34, e13.
- Lim, H.M., Park, C.J., Yook, J.H., Kim, M.S., Kim, H.Y., Hong, Y.C., 2021. The association between effort-reward imbalance, work-life balance and depressive mood in Korean wage workers: the 4th Korean Working Conditions Survey. *Ann Occup Environ Med* 33, e2.
- Malouff, J.M., Thorsteinsson, E.B., Schutte, N.S., 2005. The relationship between the five-factor model of personality and symptoms of clinical disorders: a meta-analysis. *J. Psychopathol. Behav. Assess.* 27, 101–114.
- Milner, A., Krnjacki, L., LaMontagne, A.D., 2016. Age and gender differences in the influence of social support on mental health: a longitudinal fixed-effects analysis using 13 annual waves of the HILDA cohort. *Publ. Health* 140, 172–178.
- Monreal, J.A., Salvat-Pujol, N., González-Rodríguez, A., Urretavizcaya, M., Crespo, J.M., Nadal, R., et al., 2022. The relationship between sex, personality traits, and the hypothalamic-pituitary-adrenocortical axis. *Arch. Womens Ment. Health* 25, 693–703.
- Mulder, J.D., Hamaker, E.L., 2021. Three extensions of the random intercept cross-lagged panel model. *Struct. Equ. Model.* 28, 638–648.
- Niedhammer, I., Bertrais, S., Witt, K., 2021. Psychosocial work exposures and health outcomes: a meta-review of 72 literature reviews with meta-analysis. *Scand. J. Work. Environ. Health* 47, 489–508.
- Nieuwenhuijsen, K., Bruinvels, D., Frings-Dresen, M., 2010. Psychosocial work environment and stress-related disorders, a systematic review. *Occup. Med. (Lond.)* 60, 277–286.
- Nigatu, Y.T., Wang, J., 2018. The combined effects of job demand and control, effort-reward imbalance and work-family conflicts on the risk of major depressive episode: a 4-year longitudinal study. *Occup. Environ. Med.* 75, 6–11.
- Oginska-Bulik, N., 2013. Negative and positive effects of traumatic experiences in a group of emergency service workers –the role of personal and social resources. *Med. Pr.* 64, 463–472.
- Oliveri, L.N., Awerbuch, A.W., Jarskog, L.F., Penn, D.L., Pinkham, A., Harvey, P.D., 2020. Depression predicts self assessment of social function in both patients with schizophrenia and healthy people. *Psychiatr. Res.* 284, 112681.
- Omunakwe, P.O., Nwinyokpugui, P., Adiele, K.C., 2018. Workplace interpersonal relationship and organizational productivity in deposit money banks in Port Harcourt. *Int. J. Econ. Manag. Soc. Sci.* 4, 153–177.
- Orth, U., Meier, L.L., Buhler, J.L., Dapp, L.C., Krauss, S., Messerli, D., et al., 2022. Effect size guidelines for cross-lagged effects. *Psychol. Methods*. In press.
- Oxenstierna, G., Magnusson Hanson, L.L., Widmark, M., Finnholm, K., Stenfors, C., Elofsson, S., et al., 2011. Conflicts at work—the relationship with workplace factors, work characteristics and self-rated health. *Ind. Health* 49, 501–510.
- Pallesen, S., Nielsen, M.B., Mageroy, N., Andreassen, C.S., Einarsen, S., 2017. An experimental study on the attribution of personality traits to bullies and targets in a workplace setting. *Front. Psychol.* 8, 1045.
- Park, K.O., Wilson, M.G., Lee, M.S., 2004. Effects of social support at work on depression and organizational productivity. *Am. J. Health Behav.* 28, 444–455.
- Pietrzak, R.H., Feder, A., Singh, R., Schechter, C.B., Bromet, E.J., Katz, C.L., et al., 2014. Trajectories of PTSD risk and resilience in World Trade Center responders: an 8-year prospective cohort study. *Psychol. Med.* 44, 205–219.
- Ramdas, D.L., Sbrilli, M.D., Laurent, H.K., 2022. Impact of maternal trauma-related psychopathology and life stress on HPA axis stress response. *Arch. Womens Ment. Health* 25, 121–128.
- Rosander, M., 2021. Mental health problems as a risk factor for workplace bullying: the protective effect of a well-functioning organization. *Ann Work Expo Health* 65, 1096–1106.
- Rosander, M., Salin, D., Viita, L., Blomberg, S., 2020. Gender matters: workplace bullying, gender, and mental health. *Front. Psychol.* 11, 560178.
- Rosseel, Y., 2012. lavaan: an R package for structural equation modeling. *J. Stat. Software* 48, 1–36.
- Rugulies, R., 2019. What is a psychosocial work environment? *Scand. J. Work. Environ. Health* 45, 1–6.
- Santini, Z.I., Jose, P.E., York Cornwell, E., Koyanagi, A., Nielsen, L., Hinrichsen, C., et al., 2020. Social disconnectedness, perceived isolation, and symptoms of depression and anxiety among older Americans (NSHAP): a longitudinal mediation analysis. *Lancet Public Health* 5, e62–e70.
- Shin, S., 2011. Validity Study of Short Forms of the Korean Version Center for Epidemiologic Studies Depression Scale. CES-D). Seoul National University, Seoul, Korea.
- Sorengaard, T.A., Langvik, E., 2022. The protective effect of fair and supportive leadership against burnout in police employees. *Saf Health Work* 13, 475–481.
- Stansfeld, S., Candy, B., 2006. Psychosocial work environment and mental health—a meta-analytic review. *Scand. J. Work. Environ. Health* 32, 443–462.
- Tetrick, L.E., Winslow, C.J., 2015. Workplace stress management interventions and health promotion. *Annu Rev Organ Psychol Organ Behav* 2, 583–603.
- van der Molen, H.F., Nieuwenhuijsen, K., Frings-Dresen, M.H.W., de Groene, G., 2020. Work-related psychosocial risk factors for stress-related mental disorders: an updated systematic review and meta-analysis. *BMJ Open* 10, e034849.
- Yang, F., Yang, B.X., Stone, T.E., Wang, X.Q., Zhou, Y., Zhang, J., et al., 2020. Stigma towards depression in a community-based sample in China. *Compr. Psychiatr.* 97, 152152.
- Zhang, H., Ye, Z., Tang, L., Zou, P., Du, C., Shao, J., et al., 2020. Anxiety symptoms and burnout among Chinese medical staff of intensive care unit: the moderating effect of social support. *BMC Psychiatr.* 20, 197.
- Zuelke, A.E., Roehr, S., Schroeter, M.L., Witte, A.V., Hinz, A., Engel, C., et al., 2020. Are social conflicts at work associated with depressive symptomatology? Results from the population-based LIFE-Adult-Study. *J. Occup. Med. Toxicol.* 15, 1.